



Stable, fast and high-energy-density LiCoO₂ cathode at high operation voltage enabled by glassy B₂O₃ modification



Aijun Zhou^{a,*}, Weihang Wang^a, Qin Liu^a, Yi Wang^b, Xu Yao^a, Fangzhu Qing^a, Enzhu Li^a, Tingting Yang^c, Long Zhang^c, Jingze Li^{a,**}

^a State Key Laboratory of Electronic Thin Films and Integrated Devices, School of Microelectronics and Solid-state Electronics, University of Electronic Science and Technology of China, Chengdu 610054, China

^b Beijing National Laboratory for Condensed Matter Physics, Institute of Physics, Chinese Academy of Sciences, Beijing 100190, China

^c State Key Laboratory of Metastable Materials Science and Technology, Yanshan University, Qinhuangdao 066004, China

HIGHLIGHTS

- LiCoO₂ powders are modified with glassy B₂O₃ through H₃BO₃ decomposition.
- High-voltage (4.5 V) cycling stability and rate capability are greatly improved.
- Lithium boron oxide (LBO) as major SEI part is formed on the surface after cycling.
- B₂O₃-modification mitigates high-voltage induced interfacial side reactions.
- The as-formed 3D glassy LBO enhances the kinetics of the electrode.

ARTICLE INFO

Article history:

Received 1 May 2017

Received in revised form

7 June 2017

Accepted 15 June 2017

Keywords:

Lithium-ion battery

LiCoO₂

High voltage

Cathode

Surface modification

Boron oxide

ABSTRACT

In this work, commercial LiCoO₂ is modified with a glassy B₂O₃ by solution mixing with H₃BO₃ followed by post-calcination in order to enhance its high-voltage electrochemical performance. The glassy B₂O₃ coating/additive is believed to serve as an effective physiochemical buffer and protection between LiCoO₂ and liquid electrolyte, which can suppress the high-voltage induced electrolyte decomposition and active material dissolution. During the early cycling and due to the electrochemical force, the as-coated B₂O₃ glasses which have 3D open frameworks tend to accommodate some mobile Li⁺ and form a more chemically-resistant and ion-conductive lithium boron oxide (LBO) interphase as a major component of the solid electrolyte interphase (SEI), which consequently enables much easier Li⁺ diffusion/transfer at the solid-liquid interfaces upon further cycling. Due to the synergetic effects of B₂O₃ coating/modification, the high-voltage capacity and energy density of the B₂O₃-modified LiCoO₂ cathode are promisingly improved by 35% and 30% after 100 cycles at 1 C within 3.0–4.5 V vs. Li/Li⁺. Meanwhile, the high-rate performance of the B₂O₃-modified electrode is even more greatly improved, showing a capacity of 105 mAh g⁻¹ at 10 C while the bare electrode has dropped to no more than 30 mAh g⁻¹ under this rate condition.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

The increasing demand for high-energy-density lithium-ion batteries (LIBs) has motivated huge efforts to explore high-performance electrode materials. Since commercialization of LIBs, LiCoO₂ has been dominating the cathode market due to its easy production and good cycling stability [1–4]. Though various new classes of materials have been developed in the last decades, LiCoO₂ still has vast superiority as small-format power units for portable devices considering its highest tapping density and maturity in

* Corresponding author.

** Corresponding author.

E-mail addresses: zhouaj@uestc.edu.cn (A. Zhou), 2496637025@qq.com (W. Wang), 214135373@qq.com (Q. Liu), wangyi4077@126.com (Y. Wang), 631763614@qq.com (X. Yao), qingfz@uestc.edu.cn (F. Qing), lienazhu@uestc.edu.cn (E. Li), 18712787539@163.com (T. Yang), lzhang@ysu.edu.cn (L. Zhang), lijingze@uestc.edu.cn (J. Li).

battery production among all candidates. As the base material of other layered cathodes and a classical model material for mechanism studies, LiCoO_2 has been receiving unfailing research interest in both experimental and theoretical aspects [5–11].

In recent years, the focus of research in LiCoO_2 has shifted to the application at high voltages [9], which can increase the degree of Li^+ utilization in the lattice structure and thus push up the limit of its capacity and energy density. Previously, LiCoO_2 batteries were charged to a cut-off potential no more than 4.2 V vs. Li/Li^+ , delivering only half of its theoretical capacity (274 mAh g^{-1}) and energy density ($\sim 1000 \text{ Wh Kg}^{-1}$, estimated with a constant voltage of 3.9 V vs. Li anode). When the potential is elevated to 4.5 V, the initial discharge capacity and energy density of LiCoO_2 can be increased to 190 mAh g^{-1} and 700 Wh Kg^{-1} at 1 C ($\sim 70\%$ utilization), which is very promising for next-generation applications. Unfortunately, pure LiCoO_2 is not stable and its rate capability also becomes very poor under such high-voltage conditions. Multiple reasons are believed to account for the high-voltage degradation of LiCoO_2 , such as oxidation decomposition of the liquid electrolytes, structure collapse, Co^{4+} dissolution and oxygen release from the active materials, and fast accumulation of the so-called solid electrolyte interphase (SEI) which is electrochemically irreversible and kinetically unfavorable [12–14].

To address the high-voltage issues of LiCoO_2 , numerous attempts have been devoted which can be classified into three strategies: 1) using additives in electrolytes [15,16], 2) lattice element doping [11,17] and 3) surface coating/modification with another active/inactive materials [9,18–20]. In the aspect of surface coating, particular attention has been paid to the use of some large band-gap oxides such as Al_2O_3 [21–23], ZrO_2 [24], TiO_2 [22,25] and ZnO [26–28]. Take the most attractive Al_2O_3 coating that has been universally applied for many other electrode materials for example, the improvements of the cycling stability and rate performance of LiCoO_2 are mainly ascribed to the formation of a LiAlO_2 - LiCoO_2 solid solution on the particle surface which can release elastic strains, provide Li^+ pathways and avoid oxygen loss from LiCoO_2 [29–32]. In addition to the use of metal oxides, some solid electrolyte materials (e.g. Li_2CO_3 [33], Li_3PO_4 [34], LiPON [35] and $\text{Li}_4\text{Ti}_5\text{O}_{12}$ [36]) in polycrystalline or amorphous forms have also been used to modify LiCoO_2 taking advantages of their high electrochemical stability and ion-conducting property, the latter is supposed to be important to enhance the interfacial kinetics of LiCoO_2 electrode.

Compared with the numerous studies on metal oxide coatings, there are few reports regarding the use of non-metallic oxides for surface coating/modification of LiCoO_2 . B_2O_3 is an interesting candidate for study since it comes from the same group of Al_2O_3 and similarly has a large band-gap and high chemical stability in acid condition due to strong B–O bonding. B_2O_3 is normally known as a glassy material with 3D networks allowing the conduction of ions in it. This property makes it a very important solid electrolyte material, e.g. as a glass former in Li_2O - B_2O_3 - P_2O_5 [37,38] or as an ion-conduction promoter in $\text{LiTi}_2(\text{PO}_4)_3$ [39] and LiTiO_2 [40]. In the aspect of surface modification, Su et al. [41] reported effectively restricted sulfur or iodine dissolution in B_2O_3 -modified Li–S and Li– I_2 batteries. B_2O_3 addition was also reported to be able to greatly influence the electrochemical performance of some carbon-based anodes (hard carbon [42,43], graphite [44] and graphene [45]) due to the change the electronic structure of these materials by partial B doping. Furthermore, the giant volume change in Si [46] and SnO_2 -based [47,48] anodes can be well accommodated by B_2O_3 modification thanks to the 3D glassy feature of B_2O_3 . Considering the high band-gap, high chemical stability and the 3D ion-conduction properties, B_2O_3 is supposed to be a promising surface modification material for addressing the high-voltage problems of LiCoO_2 . Previously, B_2O_3 was used to modify LiCoO_2

through the solid-state synthesis and the electrochemical performance was studied with the cut-off voltage of 4.2 V [49]. However, there has been no report in literature regarding the high-voltage ($>4.2 \text{ V}$) electrochemical performances of LiCoO_2 with B_2O_3 modification.

In this work, B_2O_3 -modified LiCoO_2 cathode materials are synthesized by a facile and scalable solution-mixing/post-calcination method using H_3BO_3 as the precursor modifier. The electrochemical performances of the B_2O_3 -modified electrodes are investigated at the cut-off potential of 4.5 V vs. Li/Li^+ . It is found that both the cycling stability and the rate capability of the electrode are significantly improved by the B_2O_3 modification. The high-voltage induced side reactions at the electrode-electrolyte interfaces are effectively mitigated due to effective physicochemical protection from the B_2O_3 coating/additive. The enhancement of the rate capability is attributed to the electrochemical formation of a special glassy lithium boron oxide (LBO) interface as the major component of the SEI, which enables easier transfer of Li^+ and reduces the interfacial resistance during the long-term high-voltage cycling.

2. Experimental

2.1. B_2O_3 -modification of LiCoO_2

1.5 g LiCoO_2 powders (99.9%, Hunan Changyuan Like Co., Ltd.) were put into a prepared transparent H_3BO_3 solution followed by strong stirring at room temperature (RT) for 1 h. The mixture was then heated to and slowly evaporated at 90°C until a viscous slurry was formed, which was then dried at 80°C in air for 12 h. Finally, the obtained product was calcined at 500°C for 12 h in air in order to decompose the H_3BO_3 . For a comparative study, 1 wt. %, 2 wt. % and 3 wt. % H_3BO_3 (with respect to the weight of LiCoO_2) was employed and the resulted samples are named as LCO/BO1, LCO/BO2 and LCO/BO3, respectively. Additional LiCoO_2 powders were treated with all the same procedures only without the addition of H_3BO_3 and the obtained sample is named as LCO-bare as the reference.

2.2. Electrode preparation and cell assembly

The bare or B_2O_3 -modified LiCoO_2 powders were thoroughly mixed with acetylene black (AB) conductive additive and polyvinylidene fluoride (PVDF) binder with a weight ratio of 8: 1: 1 in N-methyl pyrrolidone (NMP). The slurry was homogeneously spread onto a piece of clean Al foil followed by vacuum-drying at 110°C for 12 h. The electrode foil was punched into discs of 10 mm in diameter with a loading density of $\sim 1.5 \text{ mg cm}^{-2}$. For electrochemical tests, CR2032-type coin cells were assembled in an Ar-filled glove-box using Li foil as the anode. Prior to the cell assembly, the electrodes and separators (polypropylene, Celgard 2400) were dried at 110°C and 50°C overnight in vacuum, respectively. 1 M LiPF_6 dissolved in a mixture of ethylene carbonate (EC), diethyl carbonate (DEC) and dimethyl carbonate (DMC) (1:1:1, v/v) was used as the liquid electrolyte.

2.3. Electrochemical tests

Galvanostatic charge/discharge tests of the half cells were performed using a CT2001A cell test instrument (LAND Electronic Co.) within 3.0–4.5 V vs. Li/Li^+ at RT. All cells were activated by a cycle of low-rate (0.2 C, 1 C = 140 mA g^{-1}) charge/discharge before the stability test at 1 C. The rate capability was tested stepwise from 0.02 C to 10 C and back to 0.02 C again. Cyclic voltammetry (CV) tests were carried out using an Arbin BT2000 equipment with a

sweeping rate of 0.1 mV s^{-1} . Electrochemical impedance spectroscopy (EIS) measurements were performed using an electrochemical workstation (CHI660B) in the frequency range from 10^5 to 10^{-2} Hz with a perturbation voltage of 10 mV.

2.4. Material characterizations

Differential scanning calorimetry (DSC) and thermogravimetry (TG) analysis of the materials were performed with a NETZSCH STA449C calorimeter at a temperature rate of 10 K min^{-1} in air. The phase structure of the materials was determined by X-ray diffraction (XRD, Dandong Haoyuan Instrument DX2700) using $\text{Cu K}\alpha$ radiation ($\lambda = 1.54056 \text{ \AA}$). The microstructure of the powders and electrodes was observed with a field emission scanning electron microscope (FESEM, Hitachi S3400 N). Raman spectroscopy (Renishaw in Via) and X-ray photoelectron spectroscopy (XPS, Thermo Fisher 250Xi) were used to analyze the surface of the electrodes. For XPS studies, $\text{Al-K}\alpha$ (1486.6 eV) radiation was used as the primary excitation source operated at 150 W and the binding energies were calibrated using the C 1s level (284.6 eV) as an internal reference. Ar plasma etching was used to study the in-depth XPS information. The etching rate is 0.09 nm min^{-1} for the standard Ta_2O_5 reference material. For XPS study of the cycled electrodes, the cells were disassembled in the glove-box, and the retrieved electrodes were thoroughly cleaned by DMC followed by vacuum-drying for one day in order to completely remove the electrolyte residuals.

3. Results and discussion

As a pre-study, DSC and TG measurements of pure LiCoO_2 and H_3BO_3 powders were carried out and the results are shown in Fig. 1. The commercial LiCoO_2 is found to be thermally stable up to 600°C as we observe no clear thermal reaction peaks and weight loss. As to H_3BO_3 , two evident endothermic peaks (centered at 135°C and 160°C) are observed accompanied with a weight loss of 43.5% over 400°C , which is exactly consistent with theoretical decomposition loss from H_3BO_3 to B_2O_3 . Therefore, it is believed that B_2O_3 will be obtained after calcination of the $\text{LiCoO}_2/\text{H}_3\text{BO}_3$ precursors at 500°C . According to this, the weight ratio of the final B_2O_3 to LiCoO_2 is calculated to be 0.56%, 1.12% and 1.68% for LCO/BO1, LCO/BO2 and LCO/BO3, respectively.

Fig. 2 summarizes the phase structure, morphology and surface analysis results of the bare and B_2O_3 -modified LiCoO_2 powders. As shown by the XRD results (Fig. 2a), the layered structure of LiCoO_2 (ICDD NO. 75-0532, space group: R-3m) is basically unchanged after B_2O_3 modification, and no secondary phases related with B_2O_3 or H_3BO_3 can be detected, even with the maximum B_2O_3 addition (LCO/BO3). By an additional XRD study on the decomposition

product from pure H_3BO_3 , as shown in the Supporting Information (SI, Fig. 1), we observe an evident amorphous feature of the as-decomposed B_2O_3 with very limited proportion of crystalline phase as a hybrid. Therefore, it is reasonable to assume that the B_2O_3 introduced into LiCoO_2 is primarily amorphous. As seen from the FESEM images (Fig. 2c–f), the bare and B_2O_3 -modified powders show different surface morphologies. The bare powders show a clean-cut surface morphology, while the surfaces of the modified powders (LCO/BO1) seem to be coated with an additional substance which looks more ambiguous in appearance, probably due to the glassy B_2O_3 coating. From the Raman spectrum of LCO/BO1 (Fig. 2b), an additional peak is observed at $\sim 800 \text{ cm}^{-1}$, which corresponds to the symmetric stretching vibrations of the boroxole rings (B_3O_6) in a layered glass net [50–53]. The A_{1g} peak of LCO/BO1 is found to shift slightly to smaller Raman shifts, which is believed to be induced by the addition of the boron oxide which may have weak chemical interaction with LiCoO_2 and reduce its vibration energy. In addition, the XPS spectra of LCO/BO1 (Fig. 2g–k) can more clearly evidence the presence of B_2O_3 in LCO/BO1 as seen from the much stronger B 1s peak at 192.1 eV [43] and the O 1s shoulder peak at 531.7 eV [53], while other element spectra are more or less similar to the case in LCO-bare. It is noticed that C is also detected by XPS with especially stronger intensity before plasma etching. It is possible that during the calcination process, the surface of LiCoO_2 will partially react with CO_2 in the air forming a thin Li_2CO_3 on the outer surface, which has been observed in previous studies [54,55]. This can explain the presence of the higher binding energy (BE) peak in the spectra. The strong lower BE peak in the C spectra is attributed to hydrocarbon contaminants that are always present in the analysis chamber. The Co 2p spectra are very similar for the bare and modified electrodes. The major peaks at 194.8 eV and 779.8 eV are contributed by the $\text{Co } 2p^{1/2}$ and $\text{Co } 2p^{3/2}$ in LiCoO_2 . It is noted that the $\text{Co } 2p^{3/2}$ and O 1s (528.7 eV) peaks (both related with the Co–O bonding) are slightly shifted to higher BEs at the outer surface (before plasma etching), which suggests stronger Co–O bonding on the outer surface of the material. In the spectra of the B_2O_3 -modified powders, these shifts are relatively smaller due to the presence of an amorphous B_2O_3 layer on the LiCoO_2 surface.

The high-voltage electrochemical performances of the bare and B_2O_3 -modified LiCoO_2 electrodes were investigated by galvanostatic charge/discharge tests. Fig. 3 shows the electrochemical cycling performances of the electrodes charged to the cut-off potential of 4.5 V Li/Li^+ . Encouragingly, the cycling stability of the B_2O_3 -modified electrodes is obviously improved as compared to the bare electrode. At the 100th discharge at 1 C (Fig. 3a), the maximum specific capacity is obtained in LCO/BO1 reaching 153 mAh g^{-1} , which is 35% greater than that of the bare electrode (113 mAh g^{-1}). Increasing the addition of B_2O_3 seems to cause inferior electrode performance in the early cycles as seen from the lower staging capacities of LCO/BO2 and LCO/BO3. Nevertheless, as the cycling further continuous, all modified electrodes show higher discharge capacities than the bare one. At the 100th cycle, the capacity retention values of LCO/BO1, LCO/BO2 and LCO/BO3 are 84.4%, 77.2% and 70.6%, respectively, whereas it is only 62.4% for LCO-bare (Fig. 3b). From the charge/discharge profiles (Fig. 3c), one can observe clearer and more steady charge/discharge plateaus in the profiles of the modified electrodes at larger cycle numbers (50th and 100th). Due to simultaneous improvements in both capacity and average output voltage, LCO/BO1 shows the highest gravimetric energy density of 588 Wh kg^{-1} after 100 cycles (Fig. 3b), which is 30% greater than the corresponding energy density of LCO-bare (411 Wh kg^{-1}).

Fig. 4 displays the rate performances of the electrodes. In contrast to the ultrafast degradation of LCO-bare (from 180 mAh

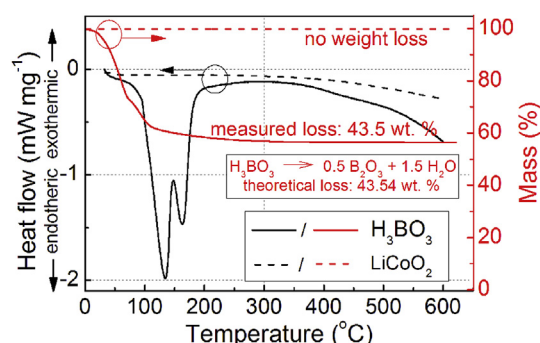


Fig. 1. DSC and TG curves of pure H_3BO_3 and the commercial LiCoO_2 powders.

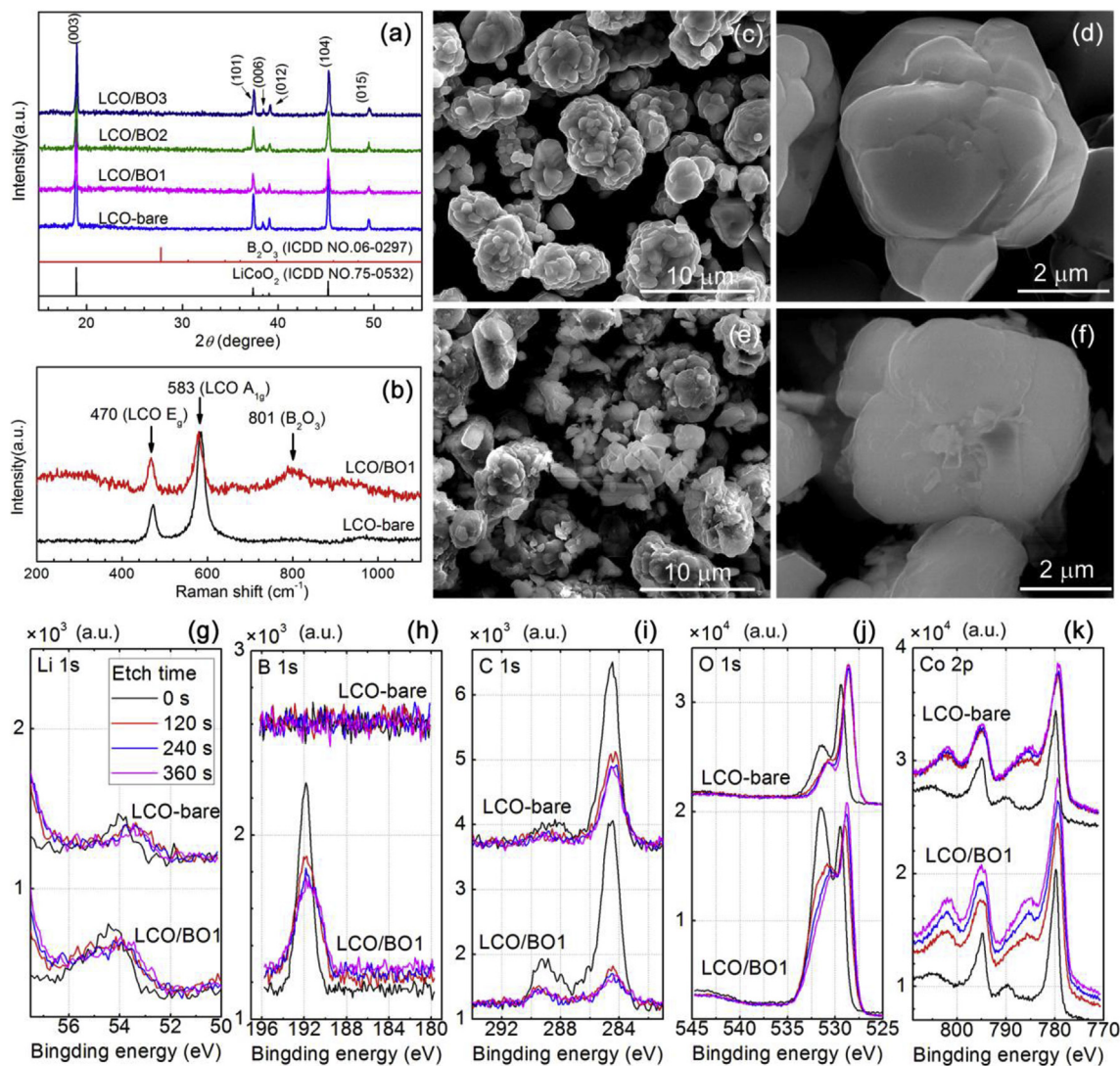


Fig. 2. Characterization results of the bare and B_2O_3 -modified $LiCoO_2$ powders: (a) XRD patterns, (b) Raman spectra, (c–f) FSEM images and (g–k) XPS spectra. Note the different magnitudes in intensity of XPS.

g^{-1} at 0.2 C to only $20 \text{ mAh } g^{-1}$ at 10 C), the B_2O_3 -modified electrodes exhibit very encouraging rate capabilities, especially for LCO/BO1 which shows the maximum capacity at the largest rate of 10 C ($95 \text{ mAh } g^{-1}$), as well as when recovered back to 0.02 C ($180 \text{ mAh } g^{-1}$) (Fig. 4a). As observed from the charge/discharge profiles at different C-rates (Fig. 4b), the bare electrode shows much stronger electrode polarization with increasing rate and its potential plateaus even disappears at 10 C. However, these unfavorable trends are remarkably suppressed in LCO/BO1. As a result, the high-rate energy density of the modified electrodes is also greatly enhanced after B_2O_3 modification (Fig. 4c). At 10 C, LCO/BO1 can still deliver an energy density of $336 \text{ Wh } kg^{-1}$, which is almost five times that of the bare electrode ($70 \text{ Wh } kg^{-1}$). The high-rate long-term stability tests were also performed to further validate the effect of B_2O_3 addition. The testing results (SI, Fig. S2) are very promising for LCO/BO1 which can deliver quite acceptable capacity ($105 \text{ mAh } g^{-1}$) and energy density ($392 \text{ Wh } kg^{-1}$) after ~ 100 cycles at a constant rate of 10 C. The electrochemical results suggest that the $LiCoO_2$ -based LIBs with B_2O_3 modifications can be more stable and faster under high-voltage conditions. The optimal H_3BO_3 addition leading to the best high-voltage performance is 1 wt. % in

this work, above which the long-term stability and rate capability are found to drop again. This can be caused by overthick coatings of B_2O_3 or its bulk precipitation in $LiCoO_2$, which may increase the polarization of the electrode due to the low electronic conductivity of B_2O_3 .

In order to understand the role of the B_2O_3 modification, the electrodes were subjected to CV studies in early cycles and EIS measurements after relatively longer cycling. From the CV profiles and the plotted ΔV (potential difference between the major cathodic and anodic peaks) (Fig. 5), it is realized that the electrode polarization is continuously increased with increasing B_2O_3 addition, which is basically in agreement with the trend observed in the early-cycle charge/discharge profiles (SI, Fig. S3). Also, the redox peaks are evidently broadened for the B_2O_3 -modified electrodes, with the hexagonal-monoclinic-hexagonal (H-M-H) phase transition peaks of $LiCoO_2$ less recognizable at the first charge. The CV results suggest that the surface potential of the electrode can be tuned by B_2O_3 coatings or additives, making the electrode surface more resistant to electrochemical oxidation at the very beginning of the cycling so that the undesired surface side reactions can be effectively mitigated. Although larger polarization is observed at

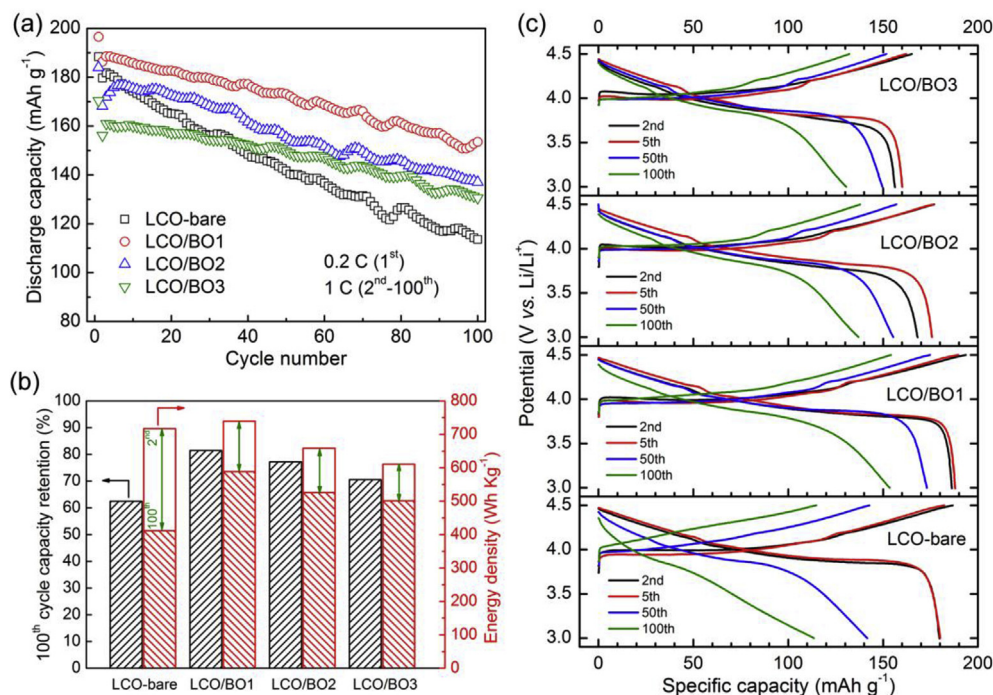


Fig. 3. Electrochemical performances of bare and B_2O_3 -modified $LiCoO_2$ electrodes in the potential range of 3.0–4.5 V vs. Li/Li^+ : (a) cycling stability, (b) 100th cycle capacity retention and gravimetric energy density and (c) selected charge-discharge profiles. The energy density of $LiCoO_2$ is calculated based on the discharge curves by integrating the discharge potential with respect to the specific capacity.

the beginning and the electrodes with excess B_2O_3 additions need more number of cycles to be fully activated (Fig. 3a), the protective role of B_2O_3 is very promising at prolonged cycling or large charge/discharge rates. From the EIS results in Fig. 6, it is found that the charge transfer resistance (R_{ct}) of LCO-bare is increased sixfold from the 5th to the 100th cycle, while the R_{ct} of LCO/BO1 is only doubled. The much slower increase of R_{ct} in LCO/BO1 indicates that after long-term cycling, Li^+ exchange and transfer in LCO/BO1 are much easier at the solid-solid (LCO-SEI) and solid-liquid (SEI-electrolyte) interfaces. Further analysis of the Warburg impedance (SI, Fig. S4) also reveals that the general diffusion of Li^+ in the electrode is evidently enhanced in LCO/BO1 after different number of cycles. These reasons can explain the excellent rate capability of LBO/BO1.

The FESEM images of the electrode surfaces after long-term cycling are displayed in Fig. 7. From the low-magnification images, no essential differences can be observed in the surface morphology of different electrodes. By a closer observation, however, one can identify a particular glassy-looking SEI layer conformably covering on the active powders in the cycled LCO/BO1. This layer should be associated with the introduced B_2O_3 coating/additive. On the bare electrode, however, the SEI seems to be formed mainly at other positions (binder or/and conductive additive part) leaving more active powders exposed to the liquid electrolyte. This will consequently result in cracks and partial corrosion/dissolution of the active material after long-term cycling at high voltage. In order to get deeper information about the SEI formed on different electrodes, XPS studies were comparatively done using the fresh and cycled electrodes, and the results are demonstrated in Fig. 8. The plots of relative atomic ratio as a function of etching time for the cycled electrodes are provided in Fig. S5 (SI), which can quantitatively compare the element differences on the surface of the bare and modified electrodes. Generally, it is observed that after 100 cycles, the surface of the B_2O_3 -modified electrode (c-LCO/BO1) shows much higher level of Li, B and O, while the bare electrode (c-LCO) is more rich in C and O. The Co element is

detected on all electrodes, but the relative peak intensity at the top surface (black lines) seems to be reduced after the modification, which should be due to the presence of an additional covering layer on the active materials. The in-depth spectra of c-LCO/BO1 reveal quite uniform distribution of B at different depths of the SEI but enriched Li on the top surface. Also, the O 1s spectra of c-LCO/BO1 are quite different with others. In contrast with the multiple peaks in c-LCO, the O spectra of c-LCO/BO1 are single-peaked and the intensity is much stronger. This difference implies that the chemical state of O on the surface of c-LCO/BO1 is essentially different with that on the surface of c-LCO. Judging from the major O peak at 531.7 eV (close to the B_2O_3 -related peak in Fig. 2h) and the unnoticeable shoulder peak at 529 eV (related with $LiCoO_2$), one can ascertain that the strong O peak of c-LCO/BO1 should originate from the boron oxide coating/additive (or its analogues). In contrast, the unmodified electrode (c-LCO) seems to be more complicated in its SEI constitutes as indicated by the much stronger surface C spectra and evidently broadened O surface spectra (535–530 eV), which imply generation of a complex organic/inorganic SEI on the bare electrode [54,56]. Furthermore, we detected much higher Li amount in the SEI of the modified electrode after cycling, while its amount on the cycled bare electrode is very limited. This information is very important because it means that the enrichment of Li element is generated after electrochemical cycling, and only on the B_2O_3 -modified electrode. From this, it is reasonable to ascertain that B_2O_3 coating layer/additive can accommodate or react with some mobile Li^+ after charge/discharge and consequently result in compositional change of the SEI at different stages of cycling.

Based on the above analysis, we can understand that the addition of B_2O_3 has not only stabilized the surface of $LiCoO_2$ at high voltage, but also enhanced the interfacial kinetics of the electrode by forming a particular SEI which is essentially different from that formed on the bare electrode. The presence of B_2O_3 is believed to have two important roles. On the one hand, it is able to alter the

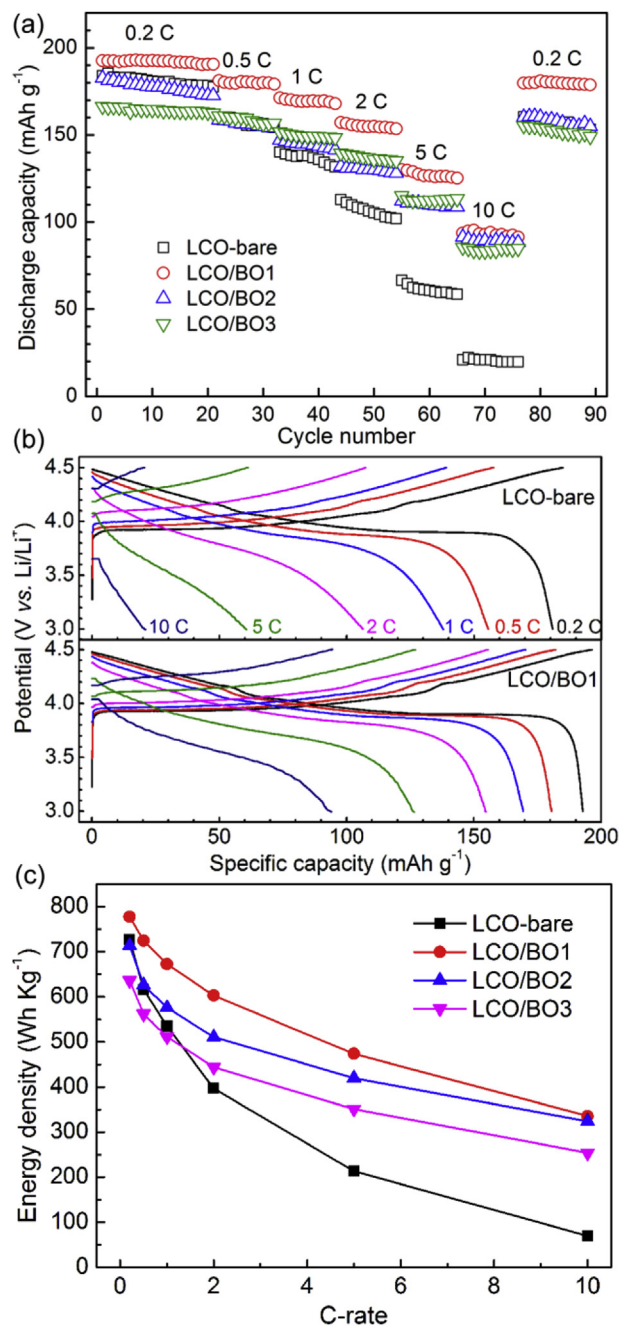


Fig. 4. (a) Rate capability of the bare and B₂O₃-modified LiCoO₂ electrodes; (b) selected charge/discharge profiles of LCO-bare and LCO/BO1; (c) calculated specific energy density at different C-rates.

surface potential of the electrode and reduce the possibility of high-voltage oxidation decomposition of the electrolyte which can cause surface dissolution of LiCoO₂ and generation of a complex organic/inorganic SEI with poor ionic conduction. On the other hand, some of the introduced B₂O₃, especially the coating layer part, is possible to undergo an electrochemical phase transition from simple boron oxide to lithium boron oxide (LBO) as indicated by the XPS results. As we know, there are many possible phases in the Li-B-O ternary system, such as LiBO₂ (Li₂O · B₂O₃), LiB₃O₅ (Li₂O · 3B₂O₃) and LiB₃O₅ (Li₂O · 2B₂O₃), which can exist in both crystalline phases and amorphous phases, the latter are known as boron compound glasses. Therefore, it is theoretically possible to form certain LBO

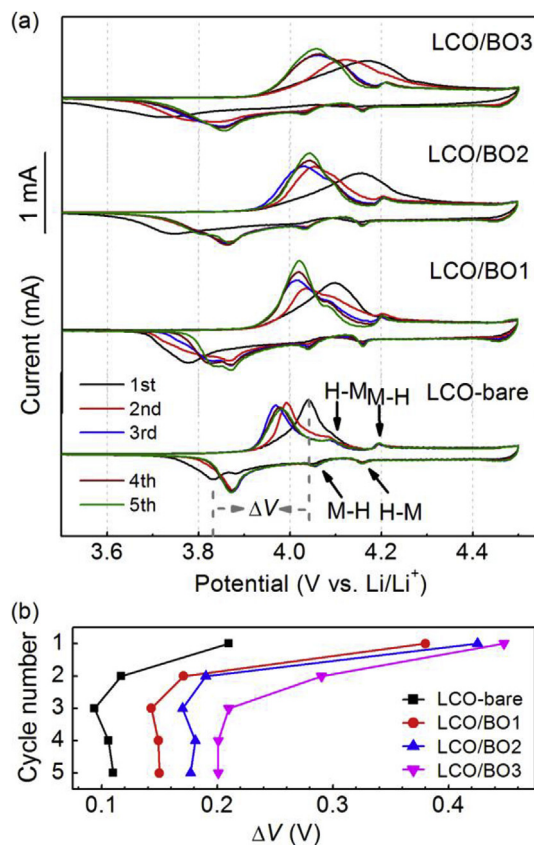


Fig. 5. (a) CV profiles of bare and B₂O₃-modified LiCoO₂ electrodes during the first five cycles; (b) variation of the potential difference between the major cathodic and anodic peaks (ΔV).

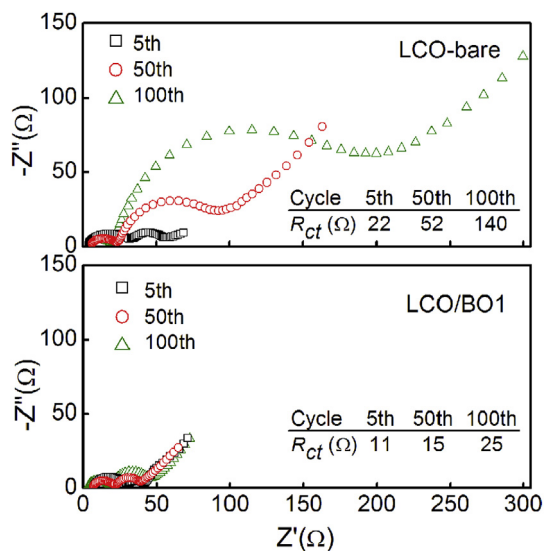


Fig. 6. EIS Nyquist plots of LCO-bare and LCO/BO1 after the 5th, 50th and 100th charge. The inset tables list the values of R_{ct} at different cycles.

phase under the electrochemical condition. In fact, there have been some previous studies employing direct LBO coating on some cathode materials (e.g. LiNi_xMn_{2-x}O₂ [57–62], LiNi_{1/3}Co_{1/3}Mn_{1/3}O₂ [63,64] and Li_{1.2}Ni_{0.2}Mn_{0.6}O₂) [65], where the LBO coating was achieved through high-temperature thermal reactions between

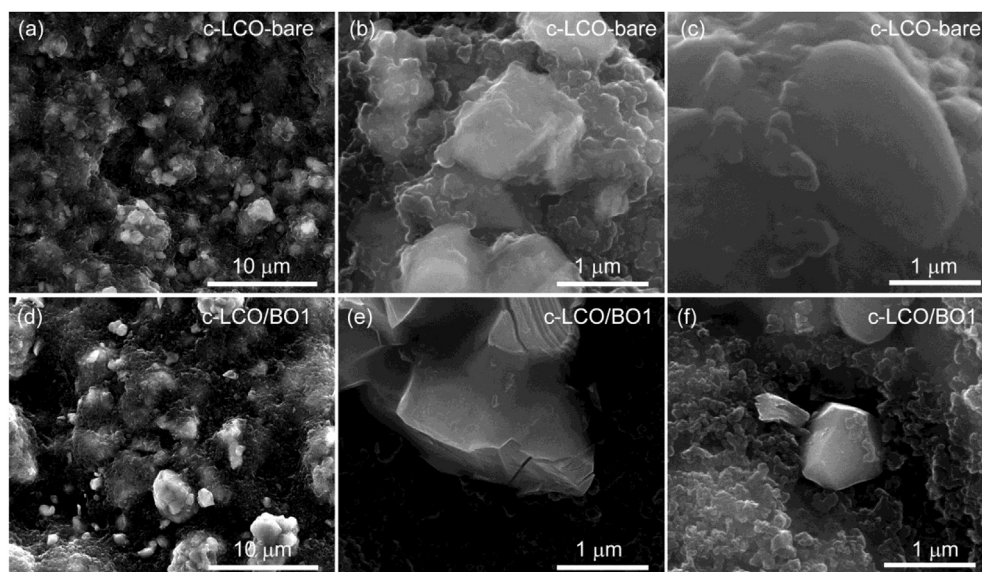


Fig. 7. FESEM images of (a–c) the bare and (d–f) B_2O_3 -modified $LiCoO_2$ electrodes after 100 cycles (the letter “c” denotes “cycled”).

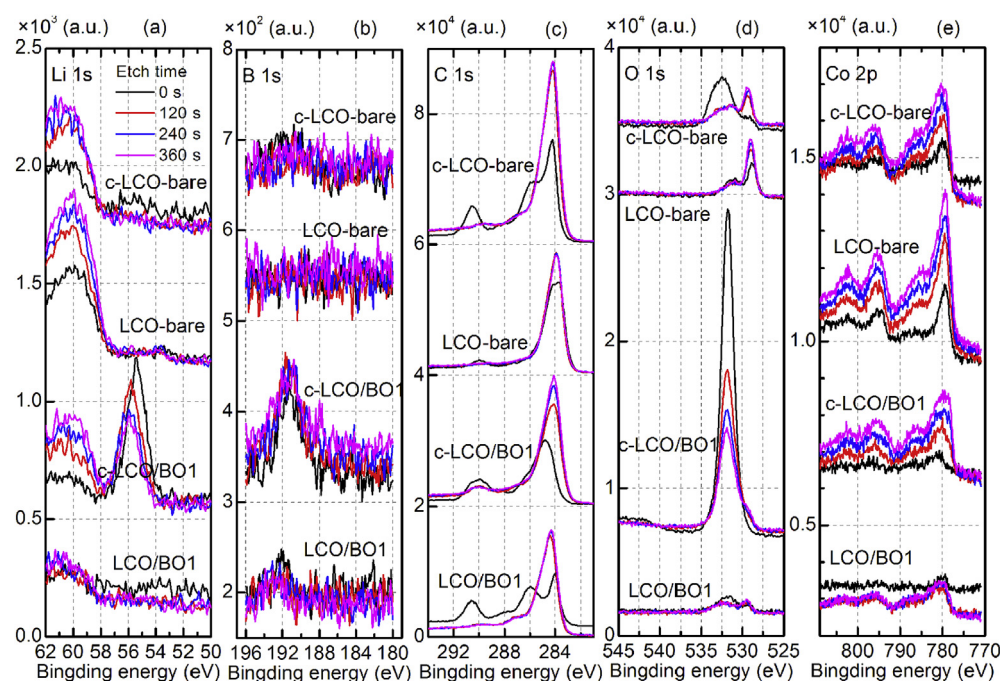


Fig. 8. XPS element spectra of the bare (LCO) and B_2O_3 -modified $LiCoO_2$ electrode (LCO/BO1) before cycling and after 100 cycles: (a) Li 1s; (b) B 1s; (c) C 1s; (d) O 1s; (e) Co 2p. Note the different magnitudes in intensity.

Li_2O (or $LiOH$) and B_2O_3 (or H_3BO_3). In this work, it is found that the simple B_2O_3 addition may similarly result in the LBO coating on the studied electrode by electrochemical reaction between B_2O_3 and Li-sources coming from the active material or electrolyte. The in-situ formed LBO interphase is believed as a beneficial SEI component due to its amorphous state and ion-conductive feature. However, excess amount of B_2O_3 addition is supposed to generate more LBO interphase which consumes active Li^+ during repeated cycling. This can be another reason that may explain the lower absolute capacities of LCO/BO2 and LCO/BO3 than LCO/BO1 as demonstrated in Fig. 3a. With a proper amount of B_2O_3 addition and consequent LBO interphase generation, the high-voltage stability and rate capability

of the $LiCoO_2$ electrode can be simultaneously improved owing to the synergetic physiochemically protection and interfacial kinetics enhancement.

4. Conclusions

A glassy boron oxide (B_2O_3) is coated on commercial $LiCoO_2$ materials by a solution-mixing/post-calcination method using H_3BO_3 as the precursor. The cycling stability and rate capability of the B_2O_3 -modified electrode are remarkably improved at a high cut-off potential of 4.5 V vs. Li/Li^+ , which is ascribed to the physiochemical protection of the B_2O_3 against electrolyte decomposition

and active material dissolution, and the formation of an ion-conductive lithium boron oxide (LBO) interface during cycling. The electrochemically generated LBO interface is the major component of the SEI on the modified electrode, which is advantageous to enhance the Li^+ diffusion and transfer at solid-liquid interfaces. In contrast, the bare electrode shows much larger charge transfer resistance after long-term cycling because of the accumulation of a complex organic/inorganic SEI which is kinetically inferior to the LBO interface. After 100 cycles, the specific capacity and energy density of the optimally modified LiCoO_2 electrode can reach 153 mAh g^{-1} and 588 Wh kg^{-1} at 1 C, and 105 mAh g^{-1} and 392 Wh kg^{-1} at 10 C, respectively, which are very promising for the development of future high-energy-density LIBs.

Acknowledgements

This work is supported by the National Science Foundation of China (11234013, 21473022, 51502032, 21673033, 51603028), the Science and Technology Bureau of Sichuan Province of China (2015HH0033, 2017HH0089) and the Fundamental Research Funds for the Central Universities of China (ZYGX2015J027).

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.jpowsour.2017.06.050>.

References

- [1] Y.K. Sun, Cycling behaviour of LiCoO_2 cathode materials prepared by PAA-assisted sol-gel method for rechargeable lithium batteries, *J. Power Sources* 83 (1999) 223–226.
- [2] H. Li, Z. Wang, L. Chen, X. Huang, Research on advanced materials for Li-ion batteries, *Adv. Mater.* 21 (2009) 4593–4607.
- [3] M.G. Kim, J. Cho, Reversible, High-Capacity Nanostructured, Electrode materials for Li-ion batteries, *J. Adv. Funct. Mater.* 19 (2009) 1497–1514.
- [4] R. Koksang, J. Barker, H. Shi, M.Y. Saidi, Cathode materials for lithium rocking chair batteries, *Solid State Ionics* 84 (1996) 1–21.
- [5] N. Wu, Y. Zhang, Y. Guo, S. Liu, H. Liu, H. Wu, Flakelike LiCoO_2 with exposed {010} facets as a stable cathode material for highly reversible lithium storage, *ACS Appl. Mater. Interfaces* 8 (2016) 2723–2731.
- [6] D.D. Liang, H.F. Xiang, X. Liang, S. Cheng, C.H. Chen, Spinel MgAl_2O_4 modification on LiCoO_2 cathode materials with the combined advantages of MgO and Al_2O_3 modifications for high-voltage lithium-ion batteries, *RSC Adv.* 7 (2017) 6809–6817.
- [7] G.M. Chen, H.L. Geng, Z.W. Wang, R.Z. Yang, Y.H. Xu, On electrochemistry of Al_2O_3 -coated LiCoO_2 composite cathode with improved cycle stability, *Ionics* 22 (2016) 629–636.
- [8] J.C. Zhang, R. Gao, L.M. Sun, H. Zhang, Z.B. Hu, X.F. Liu, Unraveling the multiple effects of Li_2ZrO_3 coating on the structural and electrochemical performances of LiCoO_2 as high-voltage cathode materials, *Electrochim. Acta* 209 (2016) 102–110.
- [9] S. Kalluri, M. Yoon, M. Jo, S. Park, S. Myeong, J. Kim, S.X. Dou, Z. Guo, J. Cho, Surface engineering strategies of layered LiCoO_2 cathode material to realize high-energy and high-voltage Li-ion cells, *Eng. Mater.* 7 (2017).
- [10] Y. Zhou, Y. Lee, H. Sun, J.M. Wallas, S.M. George, M. Xie, Coating solution for high-voltage cathode: ALF3 atomic layer deposition for freestanding LiCoO_2 electrodes with high energy density and excellent flexibility, *ACS Appl. Mater. Interfaces* 9 (2017) 9614–9619.
- [11] S. Kim, S. Choi, K. Lee, G.J. Yang, S.S. Lee, Y. Kim, Self-assembly of core-shell structures driven by low doping limit of Ti in LiCoO_2 : first-principles thermodynamic and experimental investigation, *Phys. Chem. Chem. Phys.* 19 (2017) 4104–4113.
- [12] D. Aurbach, B. Markovsky, A. Rodkin, E. Levi, Y.S. Cohen, H.J. Kim, M. Schmidt, On the capacity fading of LiCoO_2 intercalation electrodes: the effect of cycling, storage, temperature, and surface film forming additives, *Electrochim. Acta* 47 (2002) 4291–4306.
- [13] P. Arora, R.E. White, M. Doyle, Capacity Fade Mechanisms, Side Reactions, In lithium-ion batteries, *J. Electrochem. Soc.* 145 (1998) 3647–3667.
- [14] R. Yazami, Y. Ozawa, H. Gabrisch, B. Fultz, Mechanism of electrochemical performance decay in LiCoO_2 aged at high voltage, *Electrochim. Acta* 50 (2004) 385–390.
- [15] Y. Zhu, X. Luo, H. Zhi, X. Yang, L. Xing, Y. Liao, M. Xu, W. Li, Structural exfoliation of layered cathode under high voltage and its suppression by interface film derived from electrolyte additive, *ACS Appl. Mater. Interfaces* 9 (2017) 12021–12034.
- [16] M. Zhou, C. Qin, Z. Liu, Li Feng, X. Su, Y. Chen, L. Xia, Y. Xia, Z. Liu, Enhanced high voltage cyclability of LiCoO_2 cathode by adopting poly[bis-(ethoxyethoxyethoxy)phosphazene] with flame-retardant property as an electrolyte additive for lithium-ion batteries, *Appl. Surf. Sci.* 403 (2017) 260–266.
- [17] J. Yu, Z. Han, X. Hu, H. Zhan, Y. Zhou, X. Liu, Solid-state synthesis of $\text{LiCoO}_2/\text{LiCo}_{0.99}\text{Ti}_{0.01}\text{O}_2$ composite as cathode material for lithium ion batteries, *J. Power Sources* 225 (2013) 34–39.
- [18] S. Kalluri, M. Yoon, M. Jo, H.K. Liu, S.X. Dou, J. Cho, Z. Guo, Feasibility of cathode surface coating Technology for high-energy lithium-ion and beyond-lithium-ion batteries, *Adv. Mater.* (2017), <http://dx.doi.org/10.1002/adma.201605807>.
- [19] K.T. Lee, S. Jeong, J. Cho, Roles of surface chemistry on safety and electrochemistry in lithium ion batteries, *Acc. Chem. Res.* 46 (2013) 1161–1170.
- [20] L.J. Fu, H. Liu, C. Li, Y.P. Wu, E. Rahm, R. Holze, H.Q. Wu, Surface modifications of electrode materials for lithium ion batteries, *Solid State Sci.* 8 (2006) 113–128.
- [21] J. Cho, T.G. Kim, C. Kim, J.G. Lee, Y.W. Kim, B. Park, Comparison of Al_2O_3 - and AlPO_4 -coated LiCoO_2 cathode materials for a Li-ion cell, *J. Power Sources* 146 (2005) 58–64.
- [22] H.M. Cheng, F.M. Wang, J.P. Chu, R. Santhanam, J. Rick, S.C. Lo, Enhanced cycleability in lithium ion batteries: resulting from atomic layer deposition of Al_2O_3 or TiO_2 on LiCoO_2 electrodes, *J. Phys. Chem. C* 116 (2012) 7629–7637.
- [23] Y.S. Jung, P. Lu, A.S. Cavanagh, C. Ban, G.-H. Kim, S.-H. Lee, S.M. George, S.J. Harris, A.C. Dillon, Unexpected improved performance of ALD coated LiCoO_2 /graphite Li-ion batteries, *Adv. Eng. Mater.* 3 (2013) 213–219.
- [24] B.J. Hwang, C.Y. Chen, M.Y. Cheng, R. Santhanam, K. Ragavendran, Mechanism study of enhanced electrochemical performance of ZrO_2 -coated LiCoO_2 in high voltage region, *J. Power Sources* 195 (2010) 4255–4265.
- [25] A. Zhou, Y. Lu, Q. Wang, J. Xu, W. Wang, X. Dai, J. Li, Sputtering TiO_2 on LiCoO_2 composite electrodes as a simple and effective coating to enhance high-voltage cathode performance, *J. Power Sources* 346 (2017) 24–30.
- [26] W. Chang, J.-W. Choi, J.-C. Im, J.K. Lee, Effects of ZnO coating on electrochemical performance and thermal stability of LiCoO_2 as cathode material for lithium-ion batteries, *J. Power Sources* 195 (2010) 320–326.
- [27] X.Y. Dai, L.P. Wang, J. Xu, Y. Wang, A.J. Zhou, J.Z. Li, Improved electrochemical performance of LiCoO_2 electrodes with ZnO coating by radio-frequency magnetron sputtering, *ACS Appl. Mater. Interfaces* 6 (2014) 15853–15859.
- [28] X. Dai, A. Zhou, J. Xu, B. Yang, L. Wang, J. Li, Superior electrochemical performance of LiCoO_2 electrodes enabled by conductive Al_2O_3 -doped ZnO coating via magnetron sputtering, *J. Power Sources* 228 (2015) 114–122.
- [29] A. Yano, M. Shikano, A. Ueda, H. Sakaebae, Z. Ogumi, LiCoO_2 degradation behavior in the high-voltage phase transition region and improved reversibility with surface coating, *J. Electrochem. Soc.* 164 (2017) A6116–A6122.
- [30] Z. Wang, Understanding mechanism of improved electrochemical performance of surface modified LiCoO_2 , *Solid State Ionics* 175 (2004) 239–242.
- [31] J. Cho, Y.J. Kim, B. Park, Novel LiCoO_2 cathode material with Al_2O_3 coating for a Li ion cell, *Chem. Mater.* 12 (2000) 3788–3791.
- [32] Y.J. Kim, T.J. Kim, J.W. Shin, B. Park, J. Cho, The effect of Al_2O_3 coating on the cycle life performance in thin-film LiCoO_2 cathodes, *J. Electrochem. Soc.* 149 (2002) A1337–A1341.
- [33] X. Dai, A. Zhou, J. Xu, Y. Lu, L. Wang, C. Fan, J. Li, Extending the high-voltage capacity of LiCoO_2 cathode by direct coating of the composite electrode with Li_2CO_3 via magnetron sputtering, *J. Phys. Chem. C* 120 (2016) 422–430.
- [34] A. Zhou, J. Xu, X. Dai, B. Yang, Y. Lu, L. Wang, C. Fan, J. Li, Improved high-voltage and high-temperature electrochemical performances of LiCoO_2 cathode by electrode sputter-coating with Li_3PO_4 , *J. Power Sources* 322 (2016) 10–16.
- [35] Y. Kim, G.M. Veith, J. Nanda, R.R. Unocic, M.F. Chi, N.J. Dudney, High voltage stability of LiCoO_2 particles with a nano-scale Lipon coating, *Electrochim. Acta* 56 (2011) 6573–6580.
- [36] A. Zhou, X. Dai, Y. Lu, Q. Wang, M. Fu, J. Li, Enhanced interfacial kinetics and high-voltage/high-rate performance of LiCoO_2 cathode by controlled sputter-coating with a nanoscale $\text{Li}_4\text{Ti}_5\text{O}_{12}$ ionic conductor, *ACS Appl. Mater. Interfaces* 8 (2016) 34123–34131.
- [37] N.S. Saetova, A.A. Raskovalov, B.D. Antonov, T.V. Yaroslavl'tseva, O.G. Reznitskikh, N.I. Kadyrova, The influence of lithium oxide concentration on the transport properties of glasses in the $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{SiO}_2$ system, *J. Non-Cryst. Solids* 443 (2016) 75–81.
- [38] A. Mogus-Milankovic, K. Sklepik, H. Blazanovic, P. Mosner, M. Vorokhta, L. Koudelka, Influence of germanium oxide addition on the electrical properties of $\text{Li}_2\text{O}-\text{B}_2\text{O}_3-\text{P}_2\text{O}_5$ glasses, *J. Power Sources* 242 (2013) 91–98.
- [39] H. Peng, H. Xie, J.B. Goodenough, Use of B_2O_3 to improve Li-ion transport in $\text{LiTi}_2(\text{PO}_4)_3$ -based ceramics, *J. Power Sources* 197 (2012) 310–313.
- [40] M. Vijayakumar, G. Hiranikumar, M.S. Bhuvaneshwari, S. Selvasekarapandian, Influence of B_2O_3 doping on conductivity of LiTiO_2 electrode material, *J. Power Sources* 117 (2003) 143–147.
- [41] Z. Su, C.J. Tong, D.Q. He, C. Lai, L.M. Liu, C. Wang, K. Xi, Ultra-small B_2O_3 nanocrystals grown in situ on highly porous carbon microtubes for lithium-iodine and lithium-sulfur batteries, *J. Mater. Chem. A* 4 (2016) 8541–8547.
- [42] X.D. Wu, Z.X. Wang, L.Q. Chen, X.J. Huang, Increment of Li storage capacity in B_2O_3 -modified hard carbon as anode material for Li-ion batteries, *J. Electrochem. Soc.* 151 (2004) A2189–A2192.
- [43] X.D. Wu, Z.X. Wang, L.Q. Chen, X.J. Huang, Carbon/ B_2O_3 composite with higher capacity for lithium storage, *Solid State Ionics* 170 (2004) 117–121.
- [44] E. Kim, I. Oh, J. Kwak, Atomic structure of highly ordered pyrolytic graphite

- doped with boron, *Electrochem. Commun.* 3 (2001) 608–612.
- [45] L. Niu, Z. Li, W. Hong, J. Sun, Z. Wang, L. Ma, J. Wang, S. Yang, Pyrolytic synthesis of boron-doped graphene and its application as electrode material for supercapacitors, *Electrochim. Acta* 108 (2013) 666–673.
- [46] X. Wang, Z. Wen, Y. Liu, Y. Huang, T.L. Wen, Development and characterization of a novel silicon-based glassy composite as an anode material for Li-ion batteries, *Solid State Ionics* 192 (2011) 330–334.
- [47] Q. Li, J.L. Lan, Y. Liu, Y. Yu, X. Yang, Carbon nanofiber-supported B_2O_3 - SnO_x glasses as anode materials for high-performance lithium-ion batteries, *RSC Adv.* 5 (2015) 89089–89094.
- [48] R. Liu, D. Li, D. Tian, G. Xia, C. Wang, N. Xiao, N. Li, N.H. Mack, Q. Li, G. Wu, Promotional role of B_2O_3 in enhancing hollow SnO_2 anode performance for Li-ion batteries, *J. Power Sources* 251 (2014) 279–286.
- [49] S. Jouanneau, W. Bahmet, K.W. Eberman, L.J. Krause, J.R. Dahn, Effect of the sintering agent, B_2O_3 , on $LiNi_xCo_{1-2x}Mn_xO_2$ materials – density, structure, and electrochemical properties, *J. Electrochem. Soc.* 151 (2004) A1789–A1796.
- [50] W.L. Konijnendijk, J.M. Stevels, The structure of borate glasses studied by Raman scattering, *J. Non-Cryst. Solids* 18 (1975) 307–331.
- [51] K. Cho, J. Oh, T. Lee, D. Shin, Effect of P_2O_5 in Li_2O - P_2O_5 - B_2O_3 electrolyte fabricated by aerosol flame deposition, *J. Power Sources* 183 (2008) 431–435.
- [52] A. Kataev, O. Tkacheva, I. Zakiryanova, A. Apisarov, A. Dedyukhin, Y. Zaikov, Interaction of B_2O_3 with molten KF - AlF_3 and KF - NaF - AlF_3 , *J. Mol. Liq.* 231 (2017) 149–153.
- [53] X.L. Ding, Y.N. Zhou, Y. Yang, Q. Sun, F. Lu, Z.W. Fu, Electrochemical mechanism of amorphous boron oxide thin film with lithium, *Electrochem. Solid State Lett.* 14 (2011) A177–A179.
- [54] A.M. Andersson, D.P. Abraham, R. Haasch, S. MacLaren, J. Liu, K. Aminea, Surface characterization of electrodes from high power lithium-ion batteries, *J. Electrochem. Soc.* 149 (2002) A1358–A1369.
- [55] D. Aurbach, Review of selected electrode–solution interactions which determine the performance of Li and Li ion batteries, *J. Power Sources* 89 (2000) 206–218.
- [56] E. Castel, E.J. Berg, M.E. Kazzi, P. Novák, C. Villevieille, Differential electrochemical mass spectrometry study of the interface of $xLi_2MnO_3 \cdot (1-x)LiMO_2$ ($M = Ni, Co, \text{ and } Mn$) material as a positive electrode in Li-ion batteries, *Chem. Mater.* 26 (2014) 5051–5057.
- [57] H.W. Chan, J.G. Duh, J.F. Lee, Valence change by in situ XAS in surface modified $LiMn_2O_4$ for Li-ion battery, *Electrochem. Commun.* 8 (2006) 1731–1736.
- [58] A. Nagasubramanian, D.Y.W. Yu, H. Hoster, M. Srinivasan, Enhanced cycling stability of o - $LiMnO_2$ cathode modified by lithium boron oxide coating for lithium-ion batteries, *J. Solid State Electrochem.* 18 (2014) 1915–1922.
- [59] H.W. Chan, J.G. Duh, S.R. Sheen, Electrochemical performance of LBO-coated spinel lithium manganese oxide as cathode material for Li-ion battery, *Surf. Coat. Tech.* 188 (2004) 116–119.
- [60] H. Sahan, H. Goktepe, S. Patat, A. Ulgen, The effect of LBO coating method on electrochemical performance of $LiMn_2O_4$ cathode material, *Solid State Ionics* 178 (2008) 1837–1842.
- [61] H.W. Chan, J.G. Duh, S.R. Sheen, Microstructure and electrochemical properties of LBO-coated Li-excess $Li_{1+x}Mn_2O_4$ cathode material at elevated temperature for Li-ion battery, *Electrochim. Acta* 51 (2006) 3645–3651.
- [62] J.S. Chae, S.-B. Yoon, W.-S. Yoon, Y.-M. Kang, S.-M. Park, J.-W. Lee, K.C. Roh, Enhanced high-temperature cycling of Li_2O - $2B_2O_3$ -coated spinel-structured $LiNi_{0.5}Mn_{1.5}O_4$ cathode material for application to lithium-ion batteries, *J. Alloys Compd.* 601 (2014) 217–222.
- [63] S. Tan, L. Wang, L. Bian, J. Xu, W. Ren, P. Hu, A. Chang, Highly enhanced low temperature discharge capacity of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ with lithium boron oxide glass modification, *J. Power Sources* 277 (2015) 139–146.
- [64] J. Dou, X. Kang, T. Wumaier, H. Yu, N. Hua, Y. Han, G. Xu, Effect of lithium boron oxide glass coating on the electrochemical performance of $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$, *J. Solid State Electrochem.* 16 (2012) 1481–1486.
- [65] S. Chen, L. Chen, Y. Li, Y. Su, Y. Lu, L. Bao, J. Wang, M. Wang, F. Wu, Synergistic effects of stabilizing the surface structure and lowering the interface resistance in improving the low-temperature performances of layered lithium-rich materials, *ACS Appl. Mater. Interfaces* 9 (2017) 8641–8648.